

CV – Alexandre Mercier-Aubin

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Google Scholar <https://scholar.google.ca/citations?user=N3Yv5IcAAAAJ>

Personal Profile

My fields of study include computer graphics, physics-based animation, optimization and algorithm design, as well as constraint programming. They allow me to choose abstract topics while visualizing the results interactively. My work in computer graphics blends a certain artistic touch with classical science. The outcomes led to applications both in surgical simulators and in more recreational contexts such as films and video games. I also have an interest for teaching, seeing this as an important step towards the transfer of knowledge and skills to the new generation of workers and scientists.

Education

- 2020-2024** PhD in Computer Science - [McGill University](#)
Research in computer graphics on efficient animations with Prof Paul G. Kry
- 2019-2020** M. Sc. in Computer Science - [Université Laval](#)
Research in constraint programming with Prof. Claude-Guy Quimper
- 2017-2019** B. Sc. in Computer Science - [Université Laval](#)
honors
- 2014-2017** DEC in Computer Science - [Cégep Lévis-Lauzon](#)

Publications

In order to offer free and open access to scientific innovations, all of my publications are available on my website: alexandremercieraubin.com

Papers

1. **Alexandre Mercier-Aubin**, Ludwig Dumetz, Jonathan Gaudreault, and Claude-Guy Quimper. The Confidence Constraint: A Step Towards Stochastic CP Solvers. In Proceedings of the 26th International Conference on Principles and Practice of Constraint Programming (CP), pages 759-773, 2020.
2. **Alexandre Mercier-Aubin**, Jonathan Gaudreault, and Claude-Guy Quimper. Leveraging Constraint Scheduling: A Case Study to the Textile Industry. In Proceedings of the 17th International Conference on the Integration of Constraint Programming, Artificial Intelligence, and Operations Research (CPAIOR), pages 334-346, 2020.
3. **Alexandre Mercier-Aubin**, Alexandre Winter, David I. W. Levin, and Paul G. Kry. Adaptive Rigidification of Elastic Solids. In ACM Transactions on Graphics (TOG), volume 41, issue 4, article 71, 2022.
4. **Alexandre Mercier-Aubin** and Paul G. Kry. Adaptive Rigidification of Discrete Shells. In Proceedings of the ACM on Computer Graphics and Interactive Techniques (PACMCGIT), volume 6, issue 3, 2023.
5. **Alexandre Mercier-Aubin** and Paul G. Kry. A Multi-layer Solver for XPBD. In Proceedings of the Computer Graphics Forum (CGF), volume 43, issue 8, 2024.

6. Joël Pelletier-Guénette, **Alexandre Mercier-Aubin**, and Sheldon Andrews. Real-Time Triangle-SDF Continuous Collision Detection. Conditionally accepted for publication in Proceedings of the ACM on Computer Graphics and Interactive Techniques (PACMCGIT), 2025.

Workshops

7. **Alexandre Mercier-Aubin**, Jonathan Gaudreault, and Claude-Guy Quimper. Multi-Resource Scheduling with Setup Times: An Application Case to the Textile Industry. In Doctoral Program Proceedings of the 25th International Conference on Principles and Practice of Constraint Programming (CP), 2019.

PhD Thesis

8. **Alexandre Mercier-Aubin**, Adaptive Methods for Deformables, McGill University, 2024.

Master's Thesis

9. **Alexandre Mercier-Aubin**, Ordonnancement de tâches sous contraintes sur des métiers à tisser, Université Laval, 2020.

Posters

10. **Alexandre Mercier-Aubin**, Adaptive Rigidification of Elastic Solids Prototype, Graphics Interface (GI), 2022.
11. **Alexandre Mercier-Aubin**, Adaptive Rigidification of Elastic Solids Prototype, colloque REPARTI, 2022.

Talks

12. The Confidence Constraint: A Step Towards Stochastic CP Solvers. International Conference on Principles and Practice of Constraint Programming (CP), 2020.
13. Leveraging Constraint Scheduling: A Case Study to the Textile Industry. International Conference on the Integration of Constraint Programming (CPAIOR), 2020.
14. Adaptive Method for Soft Body Simulations. Tomatograph, 2021.
15. Adaptive Rigidification of Elastic Solids. Special Interest Group on Computer Graphics and Interactive Techniques (SIGGRAPH), 2022.
16. Infographie et Animation Physique : Solidification de Matériaux Viscoélastiques. Séminaire Université Laval, 2022.
17. Adaptive Rigidification of Discrete Shells. Symposium on Computer Animation (SCA), 2023.
18. Strain-based Multi-Layer solver for XPBD. Quebec-Ontario pre-SIGGRAPH (GraphQuOn), 2023.
19. A Multi-layer Solver for XPBD. Symposium on Computer Animation (SCA), 2024.

Prizes and Scholarships

- **Fonds de recherche du Québec (FRQNT): 2nd and 3rd cycle scholarship**
25 000\$ *per year, up to 3 years*
- **Natural Sciences and Engineering Research Council of Canada (NSERC) Alliance Grant**
15 000 Symgery partnership
- **Bourse de doctorat Hydro-Québec en Science**
15 000\$ *per year, up to 2 years*

- **School of Computer Science PhD funding, McGill University**
21 000\$ *per year, up to 3 years*
3 000\$ top up from Mechanical Engineering.
- **MITACS accelerate, CRISI**
39 000\$
- **Undergraduate Research Fellowship 2019-2020, Université Laval**
I declined 6500\$
- **Association for Constraint Programming, CP2019**
450\$

Research and Teaching

2025/08 Now	Sherbrooke University <i>Assistant Professor</i> Teach and do research in computer graphics / physics-based animations.
2025/01 2025/06	École de technologie supérieure <i>Postdoctoral researcher</i> Supervising graduate students, sharing research, and writing grant applications.
2024/05 - 2024/07	Autodesk <i>Research Intern</i> Rigid body differentiable simulations for surface optimization.
2020 - 2023	McGill <i>teaching assistant and AGSEM delegate</i> Hold office hours, mark exams, and present various guest lectures for the courses on computer graphics, computer animations, and introduction to computer systems.
2020	Université Laval <i>teaching assistant</i> Teach the practical aspect (weekly labs) of the advanced programming in C++ course. Mark the exams in the computer graphics course. Support students at the help centre for computer science students (CARÉ) with questions related to 15 different computer science courses.
2017/01 - 2017/08	Centre de Robotique et Vision Industrielle <i>Intern/Programmer</i> Machine learning applied to Computer Vision, program robot controllers, and develop a new website for employees.

Industry

- 2018/05 - 2018/09** - [Activision, Beenox](#)
game engine developer intern
Design and program the game engine for Call of Duty: Black Ops 4.
Technologies: DirectX, C++, LUA
Contributions: Shaders, LOD formulas, Bugfixes, HUD formulas, cross-platform compatibility, etc.
- 2016/05 - 2016/09** - [Valero, Levis](#)
computer science intern
Translate programs from Visual Basic to C#.
- 2015/05 - 2015/09** - [Consortium de ressources et d'expertises coopératives](#)
IT Technician
IT support, creating and managing a database, creating a web site, etc.

Service

- **Poster Chair, SCA, 2025**
Poster chair for the Symposium on Computer Animation conference.
- **Delegate, AGSEM, 2020-2021**
Elected as the delegate of the computer science department. As a syndicate member, one of the main mandates was to verify the tentative appointment list for the hiring unit. If the department did not respect the union's rules, then file a grievance. Other responsibilities included the organization and participation in events, formations and such.
- **V.P. Social, AGIL, 2019/9-2020/9**
Organize events for the association of graduate student in computer science at Laval University (AGIL).
- **Student volunteer, SCA, 2020 and 2024**
Support the Symposium on Computer Animation (SCA) conference by ensuring the smooth operation of sessions, main exhibitions, as well as the overall organization of activities and the design of promotional materials.
- **Chair of session, SCA, 2024**
The chair of the Physics I: Fluids, Shells and Natural Phenomena session.
- **Reviewer**
Review papers for IEEE Transactions on Visualization and Computer Graphics (TVCG), Eurographics (EG), and Motions, Interactions and Games (MIG).